Database Issues

This guide covers common database-related issues.

Symptoms

- `PrismaClientInitializationError`
- `Connection refused` errors
- `Database is out of sync`
- Slow query performance
- Migration failures

Common Issues

Prisma Client Not Generated

Symptom: `PrismaClientInitializationError: Prisma Client could not be located`

Resolution:

```
cd backend
```

```
npx prisma generate
```

Database Connection Refused

Symptom: `ECONNREFUSED` or `Connection refused`

Resolution:

1. Verify PostgreSQL is running:

   ```
   sudo systemctl status postgresql
   ```

2. Check `DATABASE_URL` in `.env`:

   ```
   DATABASE_URL="postgresql://user:password@localhost:5432/database_name?
   schema=public"
   ```

3. Verify credentials and database exist:

   ```
   psql -U user -d database_name
   ```

Schema Out of Sync

Symptom: `Migration failed to apply` or missing columns/tables

Resolution:

```
cd backend
```

```
npx prisma migrate dev
```

For production:

```
cd backend
npx prisma migrate deploy
```

Slow Queries

Diagnosis:

```
cd backend
npx prisma studio
```

Use the Query tab to test queries and check performance.

Resolution:

1. Add indexes on frequently filtered fields
2. Use `select` to limit fetched columns
3. Use `include` selectively
4. Avoid `findMany` without filters on large tables

Duplicate Ticket Numbers

Symptom: Two tickets have the same `ticketNumber`

Resolution:

The system uses a mutex to prevent this under concurrent load. If duplicates occur:

1. Check mutex implementation in `services/`
2. Ensure the server is running as a single instance (not multiple replicas without distributed locking)

Disk Space

Symptom: `No space left on device`

Resolution:

```
# Check disk usage
df -h

# Clean up old uploads
find ./uploads -type f -mtime +90 -delete

# Vacuum database
sudo -u postgres psql -d kuccps_ticketing -c "VACUUM;"
```

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